

House Price Prediction Using Linear Regression

Artificial Intelligence & Machine Learning – Task 1

1. Introduction

Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed. One of the most common and beginner-friendly machine learning algorithms is Linear Regression, which is used to predict a continuous numerical value.

The objective of this project is to build and evaluate a Linear Regression model to predict house prices using the California Housing Dataset. This task introduces the complete machine learning workflow including data loading, exploration, preprocessing, model training, evaluation, and visualization.

2. Dataset Description

The California Housing Dataset provided by scikit-learn contains housing-related information collected from California districts. The dataset is numerical, clean, and suitable for regression problems.

Features: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude
Target: MedHouseVal

3. Exploratory Data Analysis (EDA)

EDA was performed using `head()`, `info()`, and `describe()` methods to understand data structure, types, and statistics. The dataset was found to have no missing values and all numerical columns.

4. Data Preprocessing

The dataset was split into input features (X) and target variable (y). An 80-20 train-test split was applied using `train_test_split`.

5. Model Building

A Linear Regression model was implemented using `sklearn.linear_model.LinearRegression` and trained on the training dataset.

6. Model Evaluation

The model was evaluated using MAE, RMSE, and R^2 score. The obtained results show acceptable performance for a beginner-level regression task.

7. Visualization

A scatter plot of actual vs predicted house prices was created to visually analyze model performance.

8. Conclusion

The project successfully demonstrated the complete machine learning workflow. Linear Regression proved to be effective for understanding regression problems.

9. Future Improvements

Advanced models, feature scaling, hyperparameter tuning, and cross-validation can further improve performance.

10. Tools & Technologies Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn